

ABSTRACT ALGEBRA 1 - HOMEWORK 6- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

(1) **Problem 1*:**

- (a) Let $(G, \cdot, e_G)$ be a finite group and assume $|G| = n$. Show that for every element $g \in G$, $g^n = e_G$.
- (b) Let $(G, \cdot, e_G)$ be a finite group and assume $|G| = n$. Let $g \in G$, show that the order of g divides n .
- (c) Determine all possible orders of elements on Sym_5 . How many elements have order 2? How many have order 3?

- (2) **Problem 2*:(Fermat's little theorem)** Let p be a prime number. Let a be any integer, show that $a^p \equiv_p a$.

- (3) **Problem 3*:** Let a and b be non-identity elements of different orders in a group of order 155. Prove that the only subgroup of G that contains both a and b is G itself.

Problem 4*: Let $(G, \cdot)$ be a group and let $H' \subseteq H$ be subgroups of G .

- (a) (2 point) If $[G : H] = 2$ then show that H is normal in G .
- (b) (2 points) If $[G : H] = 3$ and there exists an element $x \in G \setminus H$ such that $Hx = xH$, then show that H is normal in G .